26900075 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

Regular EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBOTA01 - AGROBOTANY OF PLANTATION CROPS AND SPICES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **10** questions

Each question carries **2** marks

1. Name any two plantation crops that are export-oriented. [K] / [CO1]
2. Identify the major plantation crop in India that is important for export. [K] / [CO1]
3. Mention the role of spices in India's foreign exchange earnings. [K] / [CO1]
4. Write one medicinal use of clove. [K] / [CO2]
5. Name one tool used for propagation observed during the field visit. [K] / [CO1]
6. Summarize the watering requirements of rubber seedlings. [U] / [CO4]
7. Explain the role of improved irrigation practices in supporting high-yield varieties. [U] / [CO4]
8. Explain what you learned about seed selection and its importance in crop productivity. [U] / [CO4]

9. Name any two micronutrients required for coffee. [K] / [CO1]

10. Name two organic manures commonly applied in plantation crops like coconut and tea. [K] / [CO1]

11. Name the major insect pest responsible for boring into coffee berries. [K] / [CO3]

12. Describe berry disease in coffee plantations. [U] / [CO3]

[2x10 = 20]

Part B

Short essays

Answer any **8** questions

Each question carries **5** marks

13. Classify flower spices based on the plant part used and provide suitable examples for each category. [U] / [CO3]

14. Describe the importance of resistance to pests and diseases in coffee cultivar selection. [U] / [CO3]

15. Explain the botanical features of tea plants and their role in tea production. [U] / [CO2]

16. List the desirable characteristics of rubber cultivars [K] / [CO1]

17. List the characteristics of a good mother palm for coconut seed nut collection. [K] / [CO1]

18. Explain the process of callus induction and plant regeneration in rubber tissue culture. [U] / [CO4]

19. Explain how fertilizers are applied at different stages of tea plant growth. [U] / [CO3]

20. State the geographical and environmental conditions suitable for rubber plantation. [K] / [CO1]

21. Describe the relationship between proper postharvest handling, quality control, and shelf life of plantation crops. [U] / [CO5]

22. Differentiate between broad-spectrum and selective pesticides. [U] / [CO5]

[5x8 = 40]

Part C

Essays

Answer any **1** questions

Each question carries **10** marks

23. Analyze the nutrient management practices in coconut cultivation by comparing conventional (chemical-based) and organic (biofertilizer-based) approaches. [An] / [CO3]

24. Explain the significance of spice-based nutraceuticals in preventive healthcare with suitable examples from Cardamom, Black pepper, Cloves, and Turmeric. [U] / [CO5]

[10x1 = 10]